

UNDERGRADUATE **OPEN ENDED LAB** REPORT

Department of Mechanical Engineering

NED University of Engineering and Technology



Engineering Drawing (ME-118)

Group number: 09

Batch: 2025

Group members:

Muhammad Mustafa ME – 25222

Muhammad Taimoor Siddiqui ME – 25226

Syed Ayaan Ahmed ME – 25248

Abdullah Irfan ME – 25254

Approved by:

.....

Dr Muhammad Anique Azam

Lecturer

Course Instructor

Author's Declaration

We declare that we are the sole authors of this project. It is the actual copy of the project that was accepted by our advisor(s) including any necessary revisions. We also grant NED University of Engineering and Technology permission to reproduce and distribute electronic or paper copies of this project.

Signature and Date

Signature and Date

Signature and Date

Signature and Date

.....

.....

.....

.....

Muhammad

Muhammad

Syed Ayaan

Abdullah

Mustafa

Taimoor

Ahmed

Irfan

Siddiqui

ME – 25222

ME – 25226

ME – 25248

ME – 25254

Statement of Contributions:

Muhammad Mustafa: Drawings, Chapter 4, Chapter 5 and report compilation

Muhammad Taimoor Siddiqui: Drawings, Chapter 3 and report compilation

Syed Ayaan Ahmed: Research, Chapter 1 and report compilation

Abdullah Irfan: Research, Chapter 2 and report compilation

Acknowledgements

The making of the report and its contents was done by:

Muhammad Mustafa ME – 25222

Muhammad Taimoor Siddiqui ME – 25226

Syed Ayaan Ahmed ME – 25248

Abdullah Irfan ME – 25254

Measurement tool used: Vernier Calipers

Software(s) used: AutoCad 2024 and Microsoft Word

We would like to express our sincere gratitude to the faculty of the ICE Lab at NED University of Engineering and Technology for their support, and valuable time. Their assistance in providing the essential measurement tools and resources was instrumental in the successful completion of this report.

Chapter # 1 – Introduction

What the Assignment Is About:

The assignment requires students to create a three-dimensional (3D) model of a flange coupling using AutoCAD, a widely used computer-aided design software in engineering. The task involves applying technical drawing skills to accurately represent the components and assembly of a flange coupling in a virtual 3D environment. Through this exercise, students are expected to demonstrate proficiency in using AutoCAD's modeling tools, including extrude, revolve, and assembly features. The final model should include all essential parts of a flange coupling such as the flanges, bolts, nuts, and the shaft, assembled to reflect how the coupling functions in real-world mechanical systems.

Purpose of Studying a Flange Coupling:

Studying flange couplings is fundamental in mechanical engineering because they serve as a critical component for connecting two shafts together in power transmission applications. By learning about flange couplings, students gain insight into their design, function, and importance in machinery and industrial systems. Understanding flange couplings helps engineers ensure reliable torque transmission, alignment, and maintenance in rotating equipment. Additionally, modeling these components in AutoCAD develops practical design skills and prepares students for real-life engineering challenges, such as designing, analyzing, and maintaining mechanical connections in various industries.

What the Report Will Cover:

This report will detail the complete design process of the 3D flange coupling, including the selection and justification of dimensions for each part. It will present step-by-step AutoCAD drawings that illustrate the modeling workflow, from initial sketches to the final assembled structure. The documentation will also include technical diagrams, annotations, and screenshots to clearly demonstrate how the flange coupling was constructed and assembled in the virtual environment.

Additionally, the report will discuss the challenges faced during the modeling process and how they were addressed, ensuring a comprehensive understanding of both the theory and practical aspects of flange coupling design.

Scope:

In this assignment, we will utilize orthographic and isometric views to represent the 3D model of the flange coupling. While there are other visualization methods available such as perspective projection, which provides a more realistic sense of depth but our focus will remain on orthographic and isometric projections. This decision aligns with the requirements and learning objectives of our course, ensuring that we develop a solid understanding of technical drawing conventions commonly used in engineering design. By limiting our scope to these two projection techniques, we maintain consistency with the curriculum and reinforce the foundational skills necessary for mechanical drafting and modeling.

Chapter # 2 – Background, Applications & Working of Flange Coupling

2.1 Introduction:

Flange couplings are essential mechanical components used to connect two rotating shafts for efficient power transmission in industrial systems. They provide a secure, durable, and easily maintainable method of joining shafts, making them widely applicable in machinery such as pumps, compressors, turbines, and heat exchangers. Their simple yet robust design allows them to handle high torque loads while maintaining alignment and preventing slippage. Over time, flange couplings have evolved into different types to accommodate varying operational requirements, from rigid, high-precision applications to flexible systems that tolerate slight misalignments. Understanding their background, design variations, and working principles is essential for appreciating their role in modern engineering systems and for selecting the appropriate coupling for a given mechanical process. The following section explores the historical development, usage, and functional aspects of flange couplings, with specific reference to the coupling used in our OEL heat-exchanger setup.

2.2 About the Flange Coupling:

2.2.1 History / background of Flange Couplings:

Flange couplings are mechanical devices used to connect two shafts for the smooth transmission of power and torque. Their use dates to early industrial machinery where rigid, reliable shaft connections were needed for steam engines, pumps, and heavy mechanical drives. Over time, flange couplings evolved into several forms' rigid, flexible, and protected types to meet different alignment and safety requirements.

2.2.2 FLANGE COUPLING FOR OUR OEL:

The flange coupling used in our OEL system is taken from the heat exchanger connection, where it provides a strong, leak-proof joint between the exchanger

and the oil-carrying pipe. As hot oil flows through the heat exchanger and then exits through the flanged outlet, the coupling ensures proper sealing through a gasket, maintains alignment, and withstands the temperature and pressure of the oil line. Because it is a bolted and removable joint, it also allows easy maintenance or disconnection of the heat exchanger when required, making it ideal for this application.

2.2.3 Types of Flange Couplings:

1. Rigid Flange Coupling:
Provides a solid connection between two shafts with precise alignment. Suitable for systems with minimal or no misalignment.
2. Flexible Flange Coupling:
Incorporates a flexible element (such as rubber bushings or elastomeric inserts) to accommodate small angular, axial, or parallel misalignment.
3. Protected Type Flange Coupling:
Includes a protective flange or cover around the bolts to prevent accidental loosening and improve safety during rotation.

Applications in industries (pumps, conveyors, compressors, turbines):

Flange couplings are extensively used across industries wherever two rotating shafts must relate to high reliability. In pumping systems, they link motors to centrifugal or positive displacement pumps, ensuring steady torque delivery for fluid movement. In compressors, they provide a rigid and vibration-resistant connection between the driver and the compression unit, maintaining efficiency under high rotational speeds. In turbine-driven systems, such as steam or gas turbines, flange couplings transfer large amounts of torque to generators or auxiliary equipment with minimal power loss. Additionally, they are common in industrial mixers, blowers, gearboxes, and marine propulsion systems, where durable, safe, and easily maintainable shaft connections are required.

2.2.4 Why flange couplings are used:

Flange couplings are preferred due to their: • High torque transmission capability

- Simple and robust construction
 - Ease of installation and alignment checking
 - Availability in both rigid and flexible options to suit different conditions
- They provide a mechanically secure locking relationship between two shafts, ensuring stable rotation without slippage.

2.2.5 Working principle:

A flange coupling consists of two hubs, each fitted onto a shaft end. These hubs have flanges that are bolted together. When the driving shaft rotates, its hub and flange rotate along with it. The torque is passed through the flange interface via bolts, then into the second flange and hub, and finally into the driven shaft. The entire assembly works as a unified component once bolted.

How torque is transmitted:

Below is the elaboration for how torque is transmitted.

- Hubs:

Fixed to the shaft using a key and sometimes a set screw. They serve as the primary link between shaft and flange.

- Key:

Provides a positive mechanical fit that prevents slipping by transmitting torque from shaft to hub through shear forces.

- Flanges:

Form the main structural interface where the two coupling halves meet.

- Bolts:

Transfer torque across the flanges. Under load, the bolts experience primarily shear forces, allowing rotational motion to be transmitted from one side to the other. The reliability of torque transmission depends on proper bolt tightening, accurate key fitting, and correct shaft alignment.

2.3 Literature Review:

Flange couplings have long been recognized in mechanical engineering literature as one of the most reliable methods for connecting two rotating shafts to enable smooth transmission of power and torque. Their historical development dates back to early industrial machinery, where steam engines, pumps, and heavy mechanical drives required rigid and durable shaft connections. Over time, engineering research and industrial demands led to the evolution of various flange coupling types—rigid, flexible, and protected designs—each developed to handle different levels of misalignment, vibration, and safety requirements.

Modern literature highlights the widespread use of flange couplings in diverse mechanical systems, especially where high reliability and consistent torque transmission are essential. Applications in pumps, compressors, conveyors, turbines, and marine machinery demonstrate their importance in maintaining alignment under operational loads and ensuring minimal power loss. In systems handling fluids under pressure—such as heat exchangers—flange couplings are also valued for their ability to form strong, leak-proof joints using gaskets and bolted connections. This is especially relevant in the OEL system referenced, where the coupling withstands high temperature oil flow while allowing convenient maintenance due to its removable design.

Scholarly discussions also emphasize the mechanical principles underlying flange coupling performance. Torque transmission depends on the interaction of hubs, keys, flanges, and bolts, with bolt shear strength and shaft–hub fit being critical for reliability. Studies consistently underline the advantages of flange couplings, including robustness, ease of installation, and adaptability to alignment conditions. Collectively, existing literature affirms the flange coupling as a fundamental and enduring component in mechanical power transmission.

Chapter # 3 – Engineering Standards & Design Basis

3.1 Introduction:

Engineering components such as flange couplings must be designed and documented according to recognized engineering standards. These standards ensure that the component can be manufactured correctly, interpreted universally, and assembled reliably with mating parts. This chapter outlines the drafting, dimensional, and design principles followed during the preparation of the flange coupling model and drawings for the OEL project. The purpose of this chapter is to identify the standards that guided the preparation of the flange coupling drawing and to explain the basis behind important design decisions. Following standards ensures that the drawing is consistent with industrial expectations regarding accuracy, clarity, and compatibility. Although this project is academic, the principles used align with real-world engineering practice, enabling the design to be interpreted by any technician, machinist, or engineer familiar with ISO-based drawing conventions.

3.2 Standards:

3.2.1 Applicable Engineering Standards:

The drawing of the flange coupling was prepared according to globally recognized ISO drafting standards, which govern how technical drawings should be constructed and presented.

3.2.2 ISO 128 – Technical Drawing Conventions:

This standard defines representation rules such as the use of visible lines, hidden lines, centerlines, hatching in sectional views, and line thickness hierarchy. These conventions ensure clear communication and prevent ambiguity in manufacturing drawings.

3.2.3 ISO 129 – Dimensioning Principles:

ISO 129 governs how dimensions are placed and interpreted, including arrowheads, extension lines, placement of numerical values, and the alignment of dimension text. This ensures that all dimensions on the flange coupling are readable and follow accepted engineering practice.

3.2.4 ISO 5455 – Scales for Technical Drawings:

The drawing was prepared on an A3 sheet, and the scale was selected to allow all necessary orthographic and sectional views to fit comfortably while maintaining clarity. The selected scale ensures that the sizes of features appear proportional and that manufacturing personnel can interpret the drawing without distortion.

3.2.5 ISO 5456 – Projection Methods:

The model was drawn using 3rd Angle Projection, which is commonly used in Pakistan and is acceptable internationally. This standard governs the arrangement of views such that the top, front, and side views are positioned relative to each other in a consistent and easily interpretable manner. These standards collectively define how the drawing communicates shape, size, views, and construction details.

3.2.6 Flange and Fastener Standards:

The bolt hole sizes, pitch circle diameter (PCD), and flange thickness were selected based on standard proportions typically used in flange couplings.

3.2.7 Functional Requirements:

The design of the flange coupling is based on the operational requirements of the heat-exchanger piping system used in the OEL setup. The coupling must maintain a tight, leak-proof seal through a gasket, ensuring that hot oil does not escape under operating temperature and pressure. It must also preserve accurate alignment between the heat exchanger and the connected pipe to allow smooth flow and prevent mechanical stresses. Sufficient bolt strength and flange thickness are necessary to withstand internal loading, vibration, and thermal effects. Additionally, the coupling must remain removable to allow periodic maintenance or replacement of the exchanger without damaging the piping. These functional needs guided the choice of dimensions such as hub diameter, flange thickness, bolt size, and bolt circle diameter.

3.2.8 Reliability Considerations:

Certain critical proportions—such as the bolt circle diameter, the flange thickness, and the hub size— were kept consistent with commonly used industrial practice. These ensure that the coupling would have adequate strength, even though stress analysis was not part of the project.

Drawing Scale, Views, and Representation Standards:

The drawing was prepared using:

A3 sheet size

3rd Angle Projection (ISO 5456)

Orthographic views (Front, Top, Side)

ISO 128 line conventions for visible edges, hidden edges, and centerlines

Standard metric dimensions placed according to ISO 129

Scaling ensured that all views fit clearly on the sheet without crowding, maintaining readability and professional presentation.

3.3 Literature Review:

Engineering literature consistently emphasizes the importance of adhering to internationally recognized drafting and design standards to ensure accuracy, clarity, and universal understanding of technical drawings. The ISO system of standards forms the foundation of modern engineering documentation, particularly for mechanical components such as flange couplings. ISO 128, which defines technical drawing conventions, provides rules for line types, line hierarchy, hatching, and representation practices. These conventions eliminate ambiguity and allow manufacturers to interpret sectional and orthographic views with consistency. ISO 129, covering dimensioning principles, establishes how numerical values, extension lines, arrowheads, and orientation of text must be presented, ensuring that dimensions are communicated precisely and without misinterpretation.

Projection and scaling standards are equally critical. ISO 5455 specifies the appropriate scales for engineering drawings, enabling designers to fit multiple views on standardized sheet sizes—such as A3—while preserving clarity and proportionality. ISO 5456, which governs projection methods, standardizes the widely used 3rd Angle Projection method, ensuring consistent view placement and improving interpretability across global engineering environments.

Scholarly and industrial references also stress the need to align component dimensions with functional requirements. For flange couplings, standards relating to bolt sizes, flange thickness, hub proportions, and bolt circle diameters ensure structural reliability and compatibility with fasteners. Literature further highlights that the coupling must maintain sealing integrity, withstand mechanical loads, and allow disassembly for maintenance—criteria that directly influence design choices. Thus, the use of ISO drafting standards, combined with industry-validated

flange proportions, provides a robust framework for producing accurate, functional, and universally readable technical drawings.

Chapter # 4 – Dimensions, Tolerances & Design Explanation

4.1 Introduction:

This chapter presents the dimensional framework, tolerance selection, and detailed design rationale used in developing the flange coupling for the heat exchanger piping system. Accurate dimensions and appropriate tolerances are essential to ensure manufacturability, interchangeability, and reliable operation under thermal and mechanical loads. The dimensions were determined based on ISO drafting standards, standard fastener data, and the functional requirements of maintaining a leak-proof and mechanically stable connection. Each major feature—such as the hub, flange plate, bolt circle, and shaft connection—was sized to satisfy structural integrity, ease of assembly, and compatibility with standard workshop machining practices.

4.2 Dimensional Specification and Rationale

4.2.1 Flange Diameter and Thickness

The flange outer diameter was selected to provide sufficient room for bolt placement while maintaining adequate edge distance for structural strength. The flange thickness was chosen according to common industrial proportions, ensuring that the component can withstand internal pressure, bolt tightening forces, and thermal expansion without deformation, considering the fact it is to be used in a heat exchanger to transfer oil.

Diameter = $118.80^{+0.010}$ mm

Thickness = $13.20^{+0.004}$ mm

4.2.2 Hub Diameter and Length

The hub diameter was sized slightly larger than the connected pipe's outer diameter to allow proper machining and fit. The hub length ensures adequate surface area for shaft or pipe insertion, providing stability and proper torque transfer. This dimension also supports smooth alignment with the heat exchanger nozzle.

Outer diameter = $72.80^{+0.008}$ mm

Inner diameter = $57.10^{+0.008}$ mm

4.2.3 Pitch Circle Diameter (PCD) and Bolt Size

The bolt circle diameter was established using standard flange-coupling design practices to distribute loads evenly around the circumference. Standard M10 bolts were selected for fastening, offering sufficient clamping force to maintain a leak-proof gasket seal. The PCD ensures that bolting is symmetrical and that each fastener shares the load uniformly.

Pitch circle radius = $8.50^{+0.003}$ mm

4.2.4 Bolt Hole Diameter

The bolt holes were dimensioned slightly larger than the nominal bolt diameter to allow insertion clearance, as recommended by ISO and standard fastener tables. This prevents assembly issues while maintaining proper positioning accuracy.

Diameter = $5.00^{+0.002}$ mm

4.2.5 Gasket Seating Surface

The gasket seating width was chosen to ensure proper compression without overloading the gasket material. The surface dimensions allow uniform sealing pressure when the bolts are tightened.

4.3 Tolerances and Fits

4.3.1 General Tolerances

General tolerances for linear and angular dimensions follow ISO 2768-m (medium class), which is widely used in machined mechanical parts. This ensures manufacturability without requiring unnecessarily tight or costly tolerances.

NOTE: MEDIUM TOLERANCES WERE USED TO ENSURE ACCURACY AS WELL AS FOR COST CUTTING.

4.3.2 Hub–Shaft Fit

A close running fit was used between the hub bore and shaft to ensure correct alignment and minimize vibration. The chosen tolerance provides a balance between ease of assembly and positional accuracy.

4.3.3 Bolt Hole Tolerances

Bolt holes include clearance tolerances to accommodate standard metric bolts, allowing slight positional deviations that naturally occur during machining while still ensuring proper clamping.

4.3.4 ISO Dimension and Tolerances table:

BOLT and HEX NUT table(s):

Bolt Size	Across Flats (s)	Across Corners (e)	Head Height (k)
M6	10 mm	11.05 mm	4 mm
M8	13 mm	14.38 mm	5.3 mm
M10	17 mm	18.9 mm	6.4 mm
M12	19 mm	21.1 mm	7.5 mm
M14	22 mm	24.5 mm	8.8 mm
M16	24 mm	26.8 mm	10 mm
M20	30 mm	32.95 mm	12.5 mm
M24	36 mm	39.55 mm	15 mm

		Thread Size (d)	Pitch (P)	Nut Width Across Flats (s)	Width Across Corners (e)	Nut Thickness (m)		
		M6	1.0	10 mm	11.05 mm	5.0 mm		
		M8	1.25	13 mm	14.38 mm	6.5 mm		
		M10	1.5	17 mm	19.46 mm	8.0 mm		
		M12	1.75	19 mm	21.93 mm	10.0 mm		
		M16	2.0	24 mm	26.75 mm	13.0 mm		
		M20	2.5	30 mm	32.95 mm	16.0 mm		
		M24	3.0	36 mm	39.55 mm	19.0 mm		

Tolerance(s) table:

For length(s):

Nominal Length Range (mm)	ISO Medium (m) Tolerance (± mm)
0 – 3	± 0.1 mm
>3 – 6	± 0.1 mm
>6 – 30	± 0.2 mm
>30 – 120	± 0.3 mm
>120 – 400	± 0.5 mm
>400 – 1000	± 0.8 mm
>1000 – 2000	± 1.2 mm
>2000 – 4000	± 2.0 mm

For Nominal Diameter:

Nominal Diameter (mm)	k6 (mm)	k7 (mm)	k8 (mm)
3 – 6	+0.002 / –0.004	+0.002 / –0.006	+0.002 / –0.010
>6 – 10	+0.003 / –0.005	+0.003 / –0.008	+0.003 / –0.014
>10 – 18	+0.004 / –0.006	+0.004 / –0.010	+0.004 / –0.018
>18 – 30	+0.005 / –0.008	+0.005 / –0.012	+0.005 / –0.022
>30 – 50	+0.006 / –0.009	+0.006 / –0.015	+0.006 / –0.025
>50 – 80	+0.008 / –0.011	+0.008 / –0.018	+0.008 / –0.032
>80 – 120	+0.010 / –0.013	+0.010 / –0.021	+0.010 / –0.040
>120 – 180	+0.012 / –0.015	+0.012 / –0.025	+0.012 / –0.048

4.4 Literature Review:

The dimensional specification of a flange coupling is guided by established engineering design practices, industrial standards, and functional requirements related to torque transmission, sealing, and structural reliability. The selected flange diameter and thickness follow proportions commonly used in piping and heat-exchanger interfaces, ensuring sufficient edge distance for bolt placement and resistance against deformation during tightening or thermal loading. With an outer diameter of **118.80 mm** and thickness of **13.20 mm**, the flange provides the necessary stiffness to maintain gasket compression and prevent leakage when exposed to hot oil flow and internal pressure.

The hub dimensions were chosen to ensure proper pipe insertion, mechanical stability, and torque transfer. The hub's outer diameter of **72.80 mm** and bore diameter of **57.10 mm** align with standard machining allowances, offering an appropriate fit for the connected pipe or shaft. These dimensions ensure stable alignment with the heat-exchanger nozzle, reducing vibration and misalignment during operation.

The pitch circle dimensions for bolt placement follow standard flange design principles that promote uniform load distribution across the coupling. A bolt circle radius of **8.50 mm**, paired with **M10** bolts, provides adequate clamping force to

secure the gasket and maintain sealing integrity. Bolt holes are intentionally oversized—**5.00 mm** with controlled tolerance—to allow for clearance and compensate for minor positional variations during machining.

Tolerances applied in the design adhere to ISO frameworks, particularly **ISO 2768m**, ensuring manufacturability while avoiding the cost burden of unnecessarily tight precision. Diameter tolerances reference ISO shaft fit **k6**, allowing appropriate clearance or interference where required. Hub–shaft fits use close running tolerances to maintain alignment and minimize vibration, while bolt-hole tolerances ensure smooth assembly without compromising structural performance. These standardized tolerances and dimensional choices collectively ensure that the flange coupling remains functional, reliable, and conveniently manufacturable.

Chapter # 5 – Engineering Drawings (3D and Orthographic Views)

5.1 Introduction:

This chapter presents the engineering drawings prepared for the flange coupling, including 3D models, orthographic projections, and sectional views. These drawings were produced in accordance with internationally recognized ISO drafting standards to ensure clarity, manufacturability, and technical accuracy. The drawings serve as the primary means of communicating the geometry, dimensions, tolerances, and assembly relationships of the flange coupling to manufacturing personnel and quality inspectors. All views were generated in AutoCAD 2024, using ISO A3 sheet size, 3rd Angle Projection, and standardized line conventions.

5.2 3D Model Representation

A 3D solid model of the flange coupling was created to visualize the complete assembly and verify geometric consistency before producing 2D drawings. The model clearly shows the hub, flange plate, bolt holes, and overall profile of both mating flanges.

The 3D representation aids in checking alignment, verifying the fit of bolts, and identifying any interference between parts. It also ensures that design decisions made in Chapter 4—such as flange thickness, bolt spacing, and hub proportions—result in a structurally sound and manufacturable component.

5.3 Orthographic Views

Orthographic projections (Front, Top, and Right Side views) were generated following **ISO 5456 – Projection Methods (3rd Angle)**.

These views provide the essential dimensional information required for fabrication, including:

- Outer diameter of the flange
- Hub diameter and length
- Bolt circle diameter (PCD)
- Bolt hole spacing and diameter
- Flange thickness
- Overall alignment of features

Line types follow **ISO 128**, where visible edges, hidden edges, and centerlines are clearly distinguished. Dimensions were placed according to **ISO 129**, ensuring readability and logical placement.

BIBLIOGRAPHY

- **International Organization for Standardization (ISO)**, *ISO 128: Technical Drawings – General Principles of Presentation*, ISO, 2018.
- **International Organization for Standardization (ISO)**, *ISO 129-1: Technical Product Documentation – Dimensioning Principles*, ISO, 2018.
- **International Organization for Standardization (ISO)**, *ISO 5455: Technical Drawings – Scales*, ISO, 1999.
- **International Organization for Standardization (ISO)**, *ISO 5456: Technical Drawings – Projection Methods*, ISO, 2011.
- **International Organization for Standardization (ISO)**, *ISO 2768-1: General Tolerances – Linear and Angular Dimensions*, ISO, 1989.
- **International Organization for Standardization (ISO)**, *ISO 286-2: ISO System of Limits and Fits – Shaft and Hole Tolerances*, ISO, 2010.
- Bhandari, V. B., *Design of Machine Elements*, 4th Edition, McGraw-Hill Education, 2010.
- Juvinall, R. C., & Marshek, K. M., *Fundamentals of Machine Component Design*, 5th Edition, Wiley, 2012.
- Shigley, J. E., Mischke, C. R., & Budynas, R. G., *Mechanical Engineering Design*, 10th Edition, McGraw-Hill, 2014.
- **Machinery's Handbook**, 30th Edition, Industrial Press, 2017.
- Norton, R. L., *Machine Design: An Integrated Approach*, 6th Edition, Pearson, 2015.

4

3

2

1

D

C

B

A

D

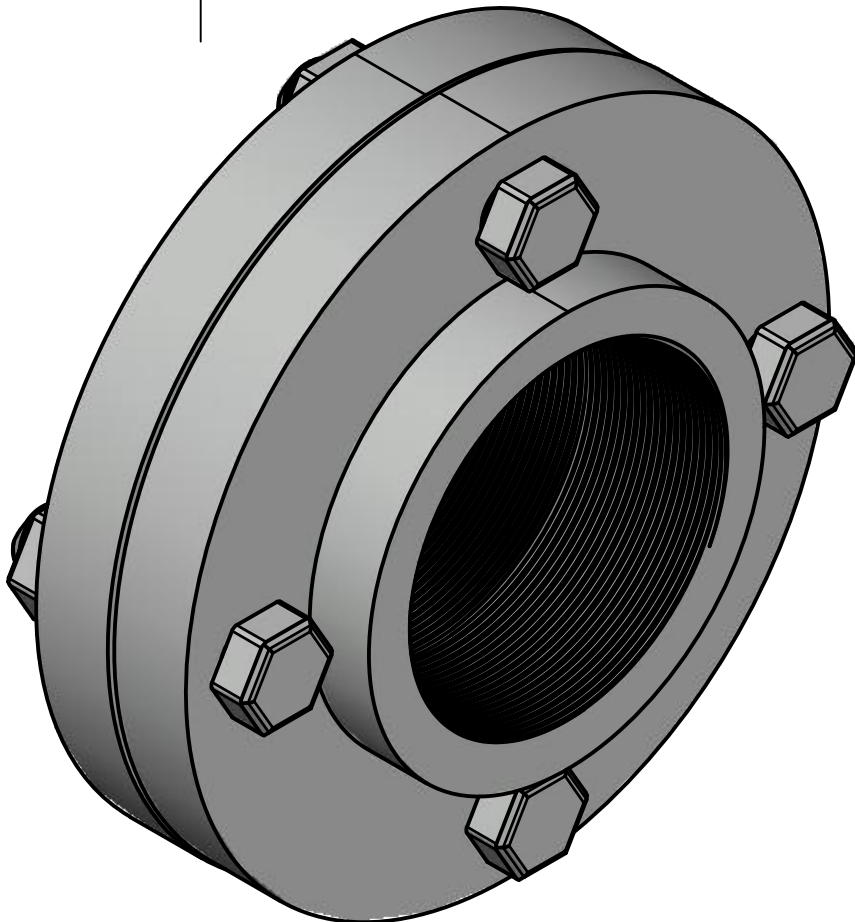
C

B

A

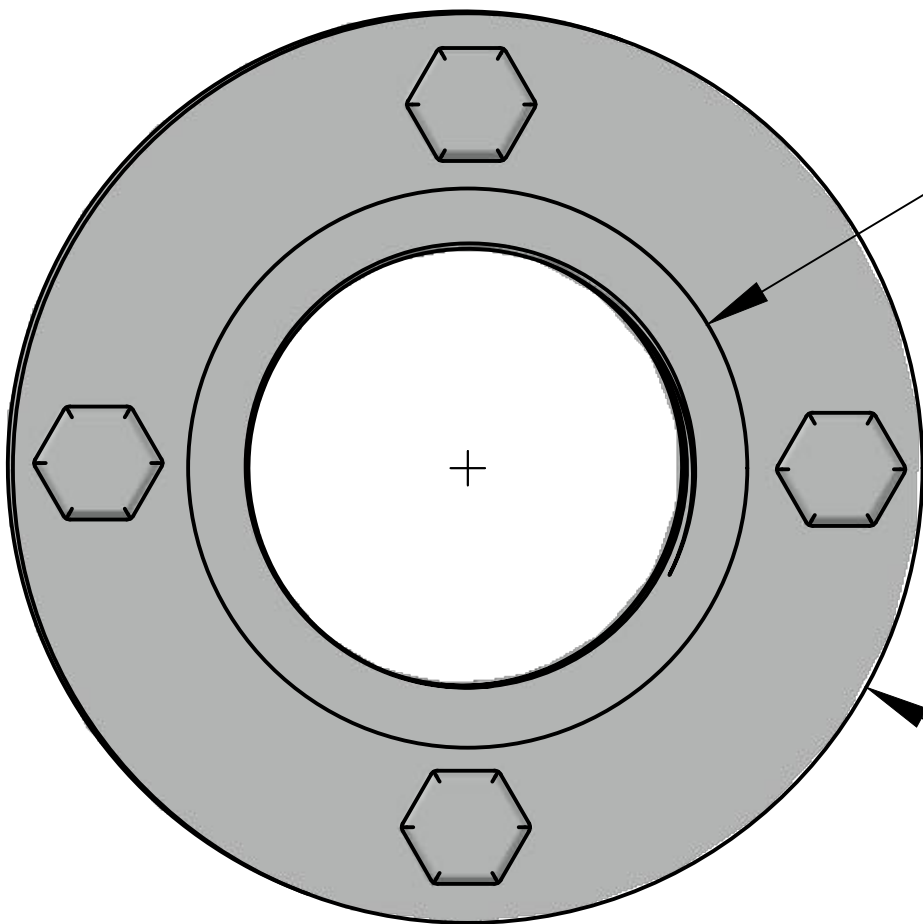


REVISIONS				
ZONE	REV	DESCRIPTION	DATE	APPROVED



48.20

26.40



Ø72.80

Ø118.40

FILE NAME				
CONTRACT NO				
DRAWN 11/26/2025				
CHECK				
APPR.				
ISSUED				
	SIZE	FSCM NO	DWG NO	REV
		—		
	SCALE 1:1	WEIGHT	SHEET	

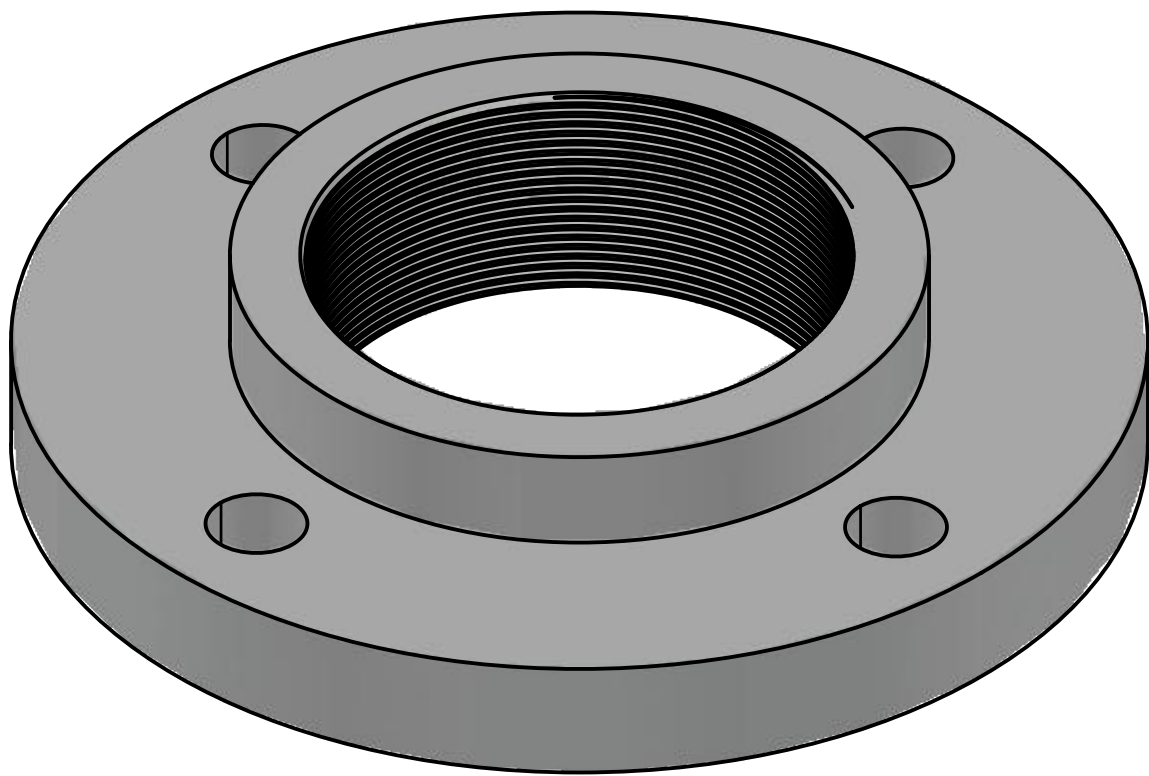
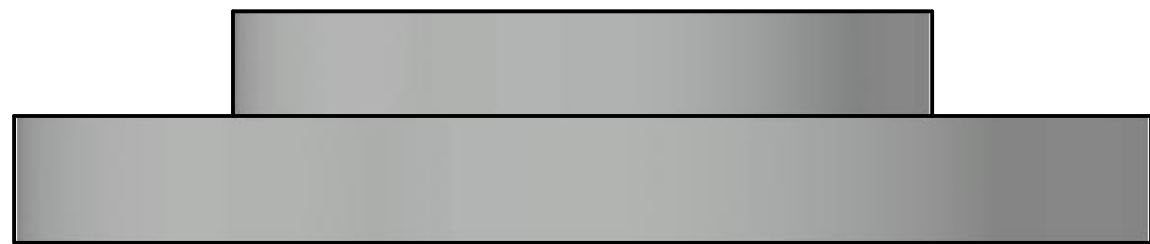
4

3

2

1

REVISIONS				
ZONE	REV	DESCRIPTION	DATE	APPROVED



D

D

C

C

B

B

A

A

Ø10.60

Ø57.10

Ø72.80

94.20

13.20

24.10

10.70

FILE NAME				
CONTRACT NO				
DRAWN 11/26/2025				
CHECK				
APPR.				
ISSUED				
	SIZE	FSCM NO	DWG NO	REV
		—		
	SCALE 1:1	WEIGHT	SHEET Flanges	

4

3

2

1

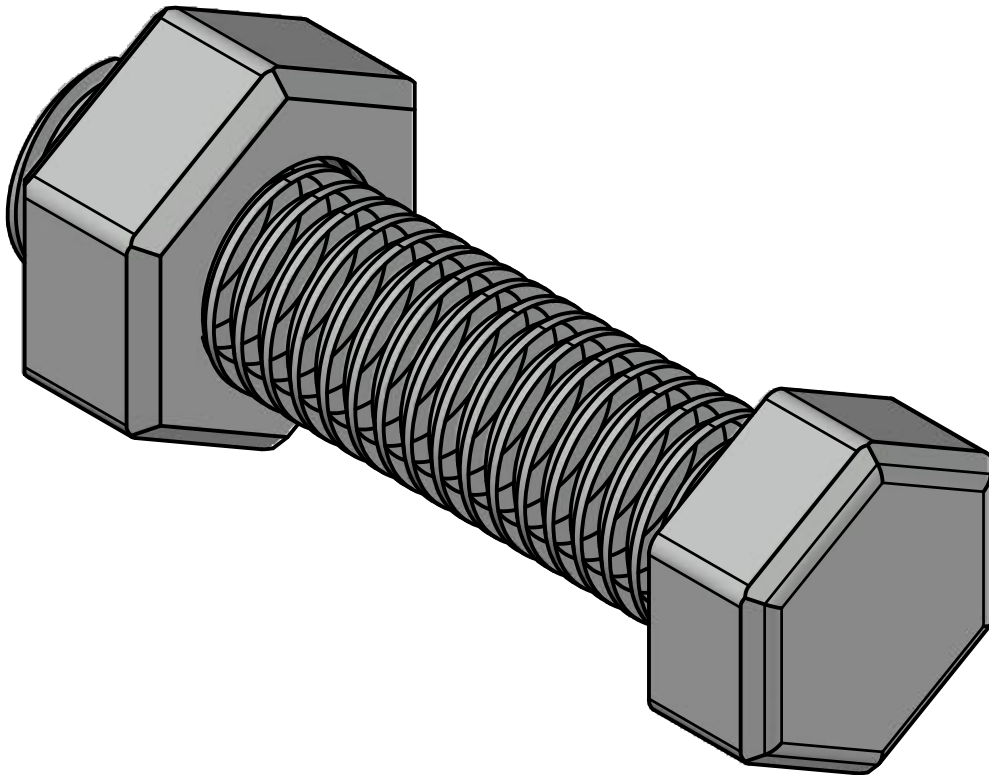
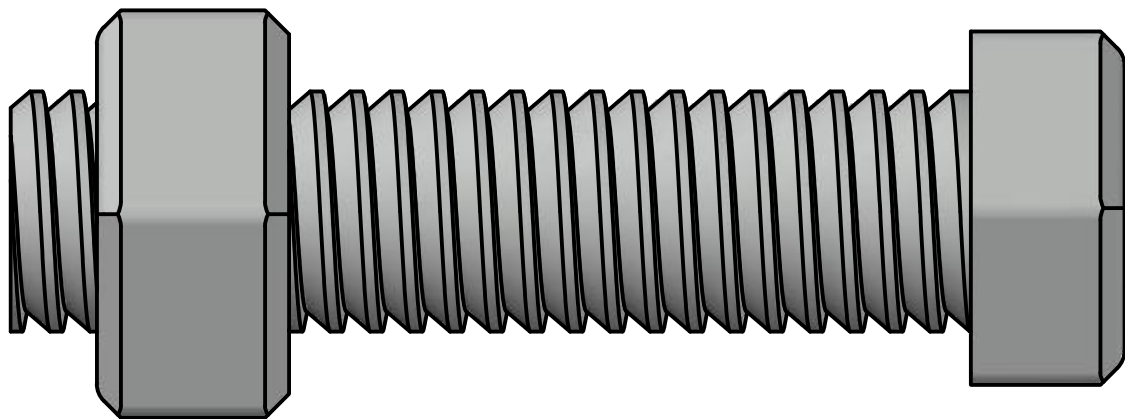
4

3

2

1

REVISIONS				
ZONE	REV	DESCRIPTION	DATE	APPROVED



40.00

10.00

R5.00

8.00

8.50

6.40

FILE NAME boltnut				
CONTRACT NO				
DRAWN 11/26/2025				
CHECK				
APPR.				
ISSUED				
	SIZE	FSCM NO	DWG NO	REV
		—		
	SCALE	1:1	WEIGHT	SHEET

4

3

2

1